6.3.8 *Fokker-Planck Equation for Langevin Dynamics*

In the special case of Langevin dynamics (6.32)–(6.33) for a system with N_c configuration variables, the density is a function of the vectors of positions $\boldsymbol{q}$, momenta $\boldsymbol{p}$, and time, and the formula for the Kolmogorov operator becomes

$$\mathcal{L}_{\mathrm{LD}}^{\dagger}\rho = -\nabla_q \cdot (\boldsymbol{M}^{-1}\boldsymbol{p}\rho) + \nabla_p \cdot ([\nabla U(\boldsymbol{q}) + \gamma \boldsymbol{p}]\,\rho) + \gamma \mathrm{k_B T}\Delta_M \rho, \tag{6.42}$$

where the mass-weighted Laplacian Δ_M is defined as

$$\Delta_M \stackrel{\mathrm{def}}{=} \nabla \cdot (\boldsymbol{M}\nabla) = \sum_{i=1}^{N_c} m_i \frac{\partial^2}{\partial p_i^2}.$$

6.4 Ergodicity of Stochastic Differential Equations

The principal question that must be answered regarding an SDE model such as Langevin dynamics has to do with its *ergodic property*. Recall from the previous chapter that we had defined ergodicity with respect to the flow map of a Hamiltonian system with compact energy surfaces as the property that temporal averages taken along trajectories of the system almost surely yield spatial averages with respect to the microcanonical distribution (alternatively, every invariant set M at energy E has probability measure either zero or one). In the context of the Gibbs-Boltzmann ensemble at given positive temperature T, we think of the phase space as being unbounded, but assume that the distribution is normalizable due to properties of the energy function and the form of the distribution.

Instead of a deterministic system of ordinary differential equations, we now have a stochastic system. Given a set of points M_0 in the phase space, we consider the stochastic paths emanating from these points, with each point from the distribution viewed as an initial condition for the evolving SDE. At any time t we take a snapshot M_t of the resulting set of random variables. Even if M_0 is bounded, the evolving set of points would be expected to expand and fill in the phase space accessible at the temperature T; eventually we hope that the points will be distributed in the entire phase space in accordance with the target measure (usually the Gibbs distribution $e^{-\beta H}$).

Equivalently we may study a single solution trajectory eminating from almost any initial point on M_0, and ask that in the long-time limit the entirety of the phase space is covered according to the desired measure. The discussion we present here is necessarily condensed; we direct the motivated reader to [269] for a more complete discussion.